AI ML ASSIGNMENT -BRAINBOT

WEEK 4 REPORT

TESTING AND FINALIZATION

The final week's objective was to thoroughly test the prototype and finalize the project for demonstration and submission. This included validating the accuracy of the outputs, polishing the user interface, and completing all required documentation.

1. Testing & Accuracy

- **What I did:** I conducted extensive testing of the chatbot's responses.
- **How I met the requirement:**
  - **Response Accuracy:** I verified that the chatbot correctly extracts locations from different phrasings (go from... to..., path between..., etc.). The backend logic consistently provides the correct and most efficient route using the A* algorithm.
  - **Navigation Path Testing:** I tested paths between all major campus locations to ensure the distances and routes were correct. The code automatically compares the performance of all four algorithms (BFS, DFS, UCS, A*) for every query, which is a form of continuous performance testing. This confirmed that the A* and UCS algorithms consistently find the optimal, shortest-distance paths.

2. UI Refinement

- **What I did:** The user interface was refined to be more user-friendly and visually appealing.
- **How I met the requirement:** I tried adding the dual-input functionality, providing both a chat input for ease and a dropdown menu for direct selection. This makes the tool more accessible to all users. I also added visual elements like the algorithm comparison cards to provide transparency and show the computational efficiency of each method.

3. Final Report & Conclusion

- **What I did:** I prepared the final project synopsis and a comprehensive report.
- **How I met the requirement:**
  - **Expected Outcome:** The expected outcome was to deliver a functional, intelligent campus navigation tool. The final product successfully meets this, providing accurate routes and a user-friendly interface.

    - **Applications:** I detailed potential applications, such as a mobile app for new students, integration into a campus website, and a foundation for a more advanced smart campus system.
    - **Conclusion:** The project successfully demonstrates the power of AI search algorithms and modern web development to solve a real-world problem.
    - **References:** I compiled a list of references, including resources on Python, the Flask framework,dialogflow,streamlit, data structures, and the A* search algorithm.

Deliverables

- **Fully Prepared Synopsis:** All sections of the final report have been completed, detailing the project's journey, methodology, and results.
- **Working Demo:** The "Bot Brain" prototype is fully functional, with a user-friendly chat interface and reliable navigation capabilities.